

IA2 - AP_DS- 2025/2026 - Tutorial 04- Essential Python Libraries: Exercices

Exercise 01: Temperature Analysis with NumPy

Problem Statement

We have the following table of daily temperatures (°C) measured over **7 days** in **3 cities**:

Day	Alger	Annaba	Oran
1	22	25	20
2	24	27	21
3	19	20	22
4	25	29	28
5	26	30	27
6	21	21	23
7	20	26	25

Tasks

1. Create a NumPy array `T` with the data above.
2. Compute and display:
 - the maximum temperature for each city,
 - the maximum temperature for each day,
 - the overall maximum temperature in the table.
3. Find the **day and city** where the overall maximum temperature occurred.
4. Create a new array showing, for each day, the **higher temperature between Alger and City Oran** (use `np.maximum`).
5. Add a **new row** representing the average temperature of each city over the 7 days.
6. Determine the **days where Annaba was the hottest city** (i.e., City Annaba's temperature exceeded both Alger and Oran).

Exercise 02: Student Grades Analysis with NumPy

Problem Statement

You are given the grades of **100 students** in **5 modules**:

Modules: `THL` , `DSS` , `DB` , `SE` , `RE`

Grades are integers between **0 and 20**. Your task is to perform statistical analysis and visualization using only **NumPy** and **Matplotlib**.

You will compute averages, variances, and standard deviations, apply boolean masks to filter grades, and visualize module comparisons using plots.

Tasks

Data Initialization

1. Generate the grades of **100 students × 5 modules** using `np.random.randint`.
2. Fix the random seed with `np.random.seed(101)` to ensure reproducibility.
3. Store the grades in a NumPy array called `grades_of_students`.
4. Define the list of modules: `['THL', 'DSS', 'DB', 'SE', 'RE']`.
5. Display a preview of the grades (first 5 students).

Basic Statistics

1. Compute the **mean grade per student**.
2. Compute the **mean grade per module**.
3. Compute the **median grade per module**.
4. Compute the **variance** and **standard deviation** for each module.
5. Find the **overall minimum and maximum grade** in the dataset.
6. Compute the 25th, 50th, and 75th percentiles for the module THL.

Boolean Mask and Filtering

1. Create a **boolean mask** indicating grades between **10 and 15**.
2. Count how many grades fall within this range.
3. Extract all grades between 10 and 15 using the mask.

Visualization (Comparison Between Modules)

1. Use a **bar plot** to compare the **average grades per module**.
2. Plot a **box plot** (or multiple histograms) to show the distribution of grades per module.
3. Customize titles, labels, and colors for clarity.

Extra practice

1. Identify students who have **all grades above 15**.
2. Compute the **top 5 students** with the highest average grades.
3. Find the module with the **highest average grade**.

Exercice 03: Hospital Data (Alger, Annaba, Oran) using Pandas

Tasks

1. **Create** a DataFrame with the following columns:
 - `City`, `Department`, `Age`, `DaysAdmitted`, `DailyCost`, `Satisfaction`, `Readmitted`

City	Department	Age	DaysAdmitted	DailyCost	Satisfaction	Readmitted
Alger	Cardiology	45	5	200	4.5	0
Annaba	Neurology	60	8	300	3.8	1
Oran	Orthopedics	30	3	150	4.2	0
Alger	Cardiology	50	7	220	4.0	0

City	Department	Age	DaysAdmitted	DailyCost	Satisfaction	Readmitted
Oran	Neurology	40	4	280	4.1	1
Annaba	Orthopedics	70	6	160	3.5	1
Alger	Cardiology	35	2	210	4.7	0
Oran	Neurology	55	9	290	3.9	1

2. Add columns:

- `TotalCost = DaysAdmitted * DailyCost`
- `AgeGroup` :
 - `"Young"` if `Age < 30`
 - `"Middle"` if `30 ≤ Age ≤ 50`
 - `"Senior"` if `Age > 50`

3. Add, Rename, and Delete Columns:

- Add a column for a 10% **Bonus** based on the daily cost
- Rename the column `DailyCost` → `DailyFee`
- Delete the column `Bonus`

4. Data Transformation:

- Create a column `HighCost` showing `"Yes"` if total cost exceeds 60,000, otherwise `"No"`
- Create a column `SatisfactionLevel` :
 - `"Excellent"` if `Satisfaction ≥ 8`
 - `"Average"` if `5 ≤ Satisfaction < 8`
 - `"Poor"` if `Satisfaction < 5`

5. Ranking and Sorting:

- Add a column ranking patients by **TotalCost** (highest rank = 1)
- Sort the dataset by **RiskScore** in descending order

6. City Summary:

For each city, calculate:

- Average total cost
- Average satisfaction
- Readmission rate (percentage of patients readmitted)

7. Risk Score Calculation:

Use the formula:

[`RiskScore = (DaysAdmitted \times 0.4) + (Age \times 0.3) + (100 - Satisfaction \times 10) \times 0.3`]
 Then classify into a new column `RiskCategory` :

- `"High"` if `RiskScore > 60`
- `"Medium"` if `40 ≤ RiskScore ≤ 60`
- `"Low"` otherwise

8. Find the city with the highest average RiskScore.

9. Apply Data Cleaning and Mapping

- Replace missing or invalid `Satisfaction` values with the mean of the column
- Map city names (`Alger` , `Annaba` , `Oran`) to region codes (e.g., `DZ-16` , `DZ-23` , `DZ-31`)
- Replace binary values in `Readmitted` column: `0` → `"No"` , `1` → `"Yes"`

10. Pivot Analysis

- Create a pivot table showing average `TotalCost` and `Satisfaction` for each city and department
- Compare the result with a grouped summary to discuss the difference between both methods

11. Export Results

- Export the main dataset to `hospital_risk.csv`
- Export the city summary to `city_summary.csv`

Exercise 04: Hospital Data Visualization — Matplotlib vs Seaborn

Matplotlib

1.Total Cost per City

- Use a **bar chart** to show total cost per city.
- Add axis labels, a title, and numerical annotations above each bar.
- Change the bar color manually (e.g., 'orange' or 'skyblue').

2.Satisfaction Distribution

- Use a **histogram** (`ax.hist`) to visualize satisfaction scores.
- Add mean and median lines with annotations.
- Adjust the number of bins to explore visual differences.

3.Cost vs Satisfaction Scatter Plot

- Plot `TotalCost` vs `Satisfaction` .
- Use `ax.scatter` with custom markers and colors.
- Highlight patients with `HighCost == "Yes"` in a different color.
- Discuss the readability of the figure.

4.Subplots by City

- Use `plt.subplots(1, 3)` to plot average satisfaction per department for each city (`Alger` , `Annaba` , `Oran`).
- Ensure consistent y-axis limits for comparison.
- Add individual titles and one global figure title.

Seaborn

1.Compare City Performances

- Create a **Seaborn barplot** showing average `TotalCost` and `Satisfaction` by `City` .
- Use `sns.barplot()` with `hue='Department'` for multi-dimensional insight.
- Discuss which city performs best economically and in satisfaction.

2.Risk Distribution

- Use `sns.boxplot()` to show the distribution of `RiskScore` across cities.
- Use `sns.violinplot()` to add richer visual detail.
- Compare both visuals and discuss what each reveals.

3. Correlations

- Use `sns.heatmap()` to visualize correlations between numeric variables.
- Comment on which factors are most related to `RiskScore` and `Readmission`.

4. Pair Relationships

- Use `sns.pairplot()` with selected variables: `Age`, `DaysAdmitted`, `TotalCost`, `RiskScore`.
- Use `hue='City'` to identify differences across cities.
- Discuss one key insight revealed by this visualization.

5. Combined dashboard

Use Seaborn to build a **combined dashboard**:

- Subplot 1: Barplot of average satisfaction per city.
- Subplot 2: Boxplot of `RiskScore` by `RiskCategory`.
- Subplot 3: Heatmap of correlations.
- Subplot 4: Countplot of readmissions (`Yes` / `No`) per city.

Add a shared title:

"Hospital Risk and Satisfaction Overview"